

The industry-agnostic compliance engine: a per-sector market analysis

Starting from a regulated-supply-chain beachhead and expanding to a cross-industry framework — with an honest reckoning of whether the white spaces are real and whether "horizontal" is a strategy or a trap.

First, the hard questions you raised — answered straight, before any market map.

"What if the white spaces are already being addressed?" Partly true. Within pharma, incumbents *are* closing the in-vertical gaps: ComplianceQuest ships grounded, cited AI; Qualifyze has a live supplier-audit network with AI risk insights; MasterControl certified to ISO 42001. If the bet were "out-feature Veeva in pharma," it would be a losing bet. **But** the six white spaces shared one trait — they are *horizontal seams*, and the deepest (a configurable, industry-agnostic engine) is the one thing no vertical incumbent will ever build, because their entire moat *is* vertical depth. Veeva becoming industry-agnostic would destroy what makes Veeva valuable. So the unoccupied space is real — but it is not "a better pharma EQMS." It is "the cross-industry compliance primitive."

"Horizontal platform" is one of the most dangerous words in enterprise software — name the trap. The graveyard is full of horizontal compliance platforms that lost to a focused vertical specialist in *every single vertical*, because regulated buyers want depth, references, and validation specific to their industry. "We do everything" reads as "best at nothing," and a generalist loses the bake-off to the specialist nearly every time. **The only horizontal strategy that works is sequenced verticalization:** win one vertical deeply enough to earn references and a validated core, then expand to the *adjacent* vertical that reuses the most architecture — never "horizontal from day one." This is exactly your instinct: *start regulated, get to agnostic*. The discipline is in the *sequence*, not the ambition.

So the reframe that makes the strategy coherent: the product is not a pharma tool and not (yet) a universal platform. It is a **regulated-supply-chain compliance engine** whose architecture happens to be industry-agnostic, deployed beachhead-first and expanded ring by ring. Competition is redefined: not Veeva (a different game), but (1) the vertical incumbent in each *new* market you enter, and (2) the horizontal platforms (ServiceNow, SAP, Salesforce, GRC tools) that could also claim the cross-industry primitive. This document maps where the engine applies, sizes it, and sequences the expansion — with the trap kept in view throughout.

Why the five-pillar engine is industry-agnostic

The engine does one universal thing: **collect** evidence → **process** it to a common model → **validate** it against whatever standard applies → **generate** the compliance/certification artifact → **seal** it as an immutable record. Strip the pharma vocabulary and that motion is identical across every standards-governed industry. What differs is configuration, not pipeline.

Pillar	The constant (the engine)	The variable (per-industry config)
1 Collect	Ingest structured + unstructured + public data + human input	Which sources (LIMS vs PLM vs sensors vs field forms)
2 Process	Normalize to a common model; resolve entities; store + index	The vocabulary (batch / lot / part / sample / shipment)
3 Validate	Compare data to a standard's controls; flag gaps; risk-weight	The standard (GMP / HACCP / IATF / AS9100 / REACH)
4 Report	Assemble cited compliance artifact; human e-sign	The artifact (batch release / CoA / PPAP / FAI / SDS)
5 Seal	Hash + append-only audit trail + AI decision record	Retention rules; which records are regulated

The macro tailwind you should know about — the standards are converging. ISO 9001 is being revised for 2026; aerospace's AS9100 is becoming IA9100 aligned to it; automotive's IATF 16949 revision (≈2027) stays structurally aligned to ISO 9001; defense adds CMMC/NIST. Industry bodies are deliberately **harmonizing quality, risk, and cybersecurity expectations across manufacturing sectors** onto a common ISO 9001 spine with vertical clause-packs on top. That is *precisely* the engine-plus-config architecture — the regulatory world is moving toward the shape the platform already has. An industry-agnostic engine with per-vertical standard packs is rowing *with* this current, not against it.

But configuration is not free. Each new vertical needs a real standards pack (the controls, encoded and maintained), new data-source connectors, domain references, and often industry-specific validation. "Configurable" lowers the cost of a new vertical from "rebuild" to "configure + validate + sell" — it does not make it zero. The strategy question is always *which* vertical is cheap enough to enter and valuable enough to justify it.

How big is the opportunity

The framework sits at the intersection of three software markets, all growing double-digit:

~\$12.5B Quality Mgmt Software (2025) → ~\$20-31B by 2030-34
~10-11% QMS market CAGR
~\$15.5B Supplier Quality Mgmt apps (2026) → ~\$42B by 2036
~13.8% Life-sciences QMS CAGR (fastest vertical)

Sources: Fortune Business Insights, Grand View, Mordor, Fact.MR, Verdantix, Emergen (2025–26). Figures vary by definition; ranges shown. QMS and SQM overlap — do not simply sum.

Where the spend concentrates (by vertical)

Vertical	Share / growth signal	What it means for entry
Discrete manufacturing (auto, aero, industrial)	Largest single block — ~32% of US QMS spend	Biggest pool, but defended by IATF/AS9100 specialists; enter via supplier-audit angle.
Healthcare / life sciences	Fastest-growing — ~13.8% CAGR	The beachhead; highest regulatory pull, highest willingness to pay.
Chemicals + Food & Beverage	83% expect bigger quality budgets (Verdantix 2025)	Strongest demand momentum; near-adjacent to the core.
Automotive (IATF 16949)	Compliance "nearly mandatory" for suppliers	Huge captive supplier base; audit redundancy is acute.

The honest read on TAM. The headline numbers (\$12.5B QMS, \$15.5B SQM) are the *total* markets — the addressable slice for a new entrant is far smaller and concentrated in the under-served edges: SMBs and emerging markets the incumbents price out, and the cross-vertical buyers no single specialist serves. Do not pitch the \$40B; pitch the specific, winnable wedge in each ring. TAM justifies the ambition; SAM/SOM justify the plan.

Sectors by architectural distance from the core

Sequence expansion by how much the engine can *reuse*, not by raw market size. The nearest ring you can win with the least new configuration is the next move.

Industry Expansion Map — by architectural distance from the regulated core

The same five-pillar engine (collect → validate vs standard → report → immutable record) applies outward in rings. Distance = how much new configuration + data each ring needs.

READING THE RINGS

Distance from core = how much NEW work to enter:
new standard pack, new data sources, new references, new validation. Nearer rings reuse more of the pharma beachhead; outer rings need more new configuration and go-to-market.

WHAT STAYS CONSTANT (the engine)

- Collect structured + unstructured data
- Validate against a standard's controls
- Generate the compliance/cert artifact
- Seal as tamper-evident immutable record
- AI assists each step, human signs

WHAT CHANGES PER INDUSTRY (config)

- The standard (GMP / HACCP / IATF / AS9100)
- The vocabulary (batch / lot / part / sample)
- The rules, thresholds, required evidence
- The report/certificate template

RING 1 · NEAR-ADJACENT
Food-Bev · Cosmetics · Blood/Tissue
batch, supplier audit, recall, lot trace

CORE
Pharma · Med Device
Nutraceuticals
(GxP, validated)

RING 2 · STANDARDS-ALIGNED
Automotive · Aerospace · Electronics
ISO 9001 spine, supplier qual, CAPA, FMEA

RING 3 · BROADER REGULATED
Chemicals · Energy · Cannabis · Agri
standards-vs-evidence, certs, custody

Sequence outward by architectural distance, not by market size alone. The nearest ring you can win with the least new work is the next move — not the biggest TAM.
Ring placement reflects architectural + regulatory similarity to the GxP core, not market size. Validate each sector's true fit before committing GTM.

The constant engine at the centre; concentric rings of industries by architectural + regulatory similarity. Inner rings reuse the most of the GxP beachhead.

Sector deep-dive (1 of 2)

Fit = architectural reuse of the engine. Pull = regulatory/buyer pressure to adopt. Density = competitive crowding.

Sector	Standard(s)	Core use case for the engine	Fit	Pull	Incumbents / whitespace
Pharma / Biotech (CORE)	GMP, 21 CFR 210/211, ICH Q7-Q10	EQMS + supplier audit; batch release; deviation→CAPA; public-data risk	Very high	Very high	Veeva, MasterControl, ComplianceQuest, Qualifyze. White: SMB/emerging-market + reproducible AI + coupling.
Medical Devices (CORE)	ISO 13485, FDA QMSR (2026), MDR	Design-history + CAPA + complaint + supplier qual; QMSR harmonization	Very high	Very high	Greenlight Guru (specialist), MasterControl. White: QMSR-transition + AI + affordability.
Nutraceuticals / Supplements (CORE)	21 CFR 111 cGMP	Same GxP motion, lighter; supplier & ingredient qualification	Very high	Med	Underserved by premium tools; price-sensitive. White: affordable GxP-grade.
Food & Beverage (RING 1)	HACCP, FDA FSMA, FSSC 22000, BRCGS	Critical-control-point monitoring, supplier audit, recall, lot traceability	High	High	SafeFood/TraceGains/SafetyChain. White: AI + supplier-network + cross-site. Strong budget momentum.
Cosmetics / Personal Care (RING 1)	ISO 22716 GMP, EU 1223/2009	GMP-lite batch + supplier + claim substantiation	High	Med	Few dedicated players; often on generic QMS. White: purpose-built + affordable.
Blood / Tissue / Cell & Gene (RING 1)	21 CFR 1271, GTP, chain-of-custody	Chain-of-custody + donor/lot traceability + release	High	High	Niche, under-digitized. White: custody + immutable record is a native strength.

Pattern in the core + Ring 1: all share the GxP "batch/lot, supplier audit, recall, validated record" shape — so the engine reuses 70–90% of its pharma configuration. These are the cheapest, fastest expansions. Food & Beverage is the standout first hop: high pull, strong budget growth, acute supplier-audit redundancy, and architecturally near-identical to pharma supplier audit.

Sector deep-dive (2 of 2)

Sector	Standard(s)	Core use case for the engine	Fit	Pull	Incumbents / whitespace
Automotive (RING 2)	IATF 16949, APQP, PPAP, FMEA	Supplier quality + PPAP/APQP evidence + audit + nonconformance	High*	Very high	Big specialists (Plex, etc.). White: AI + supplier-audit-network across the captive tier-N base.
Aerospace & Defense (RING 2)	AS9100→IA9100, NADCAP, ITAR, CMMC	Supplier qual + FAI + traceability + cyber/export controls	High*	Very high	Net-Inspect, etc. White: AS9100→ISO9001:2026→CMMC convergence + on-prem/sovereignty.
Electronics / Semiconductors (RING 2)	IPC, RoHS, REACH, conflict minerals	Supplier + substance compliance + traceability	Med-high	Med-high	Assent/iPoint (substance). White: unify quality + substance + supplier audit.
Chemicals (RING 3)	REACH, GHS, ISO 9001, Responsible Care	SDS management + supplier + substance validation + record	Med	High	Specialist EHS/SDS tools. White: quality + SDS + audit in one. 83% expect budget growth.
Energy / Oil & Gas (RING 3)	API specs, ISO 14001/45001, safety case	Asset integrity + supplier + inspection records + audit	Med-low	Med	Heavy EAM/APM incumbents. White: thinner; more new config needed.
Cannabis (RING 3)	State seed-to-sale, GACP, GPP	Seed-to-sale custody + testing + compliance record	Med	High (where legal)	Metrc/seed-to-sale niche. White: GxP-grade quality + custody; fast-growing, under-tooled.
Agriculture / Agri-food (RING 3)	GACP, GlobalG.A.P., organic certs	Farm-to-processor certification + supplier + custody	Med	Med	Fragmented. White: certification + traceability engine.

* Fit marked "High*" for auto/aero: the engine maps well, but these carry industry-specific evidence artifacts (PPAP, FAI, NADCAP) that need real configuration — more new work than Ring 1, hence the ring placement despite high architectural similarity.

Reading Rings 2-3. These are larger markets (discrete manufacturing is the biggest QMS block) but cost more to enter — industry-specific artifacts, entrenched specialists, and longer sales cycles. They are the prize, not the start. The ISO 9001:2026 convergence makes Ring 2 progressively cheaper to enter over the next few years — a reason to *position* for it now but *sequence* it after Ring 1.

How to walk outward without falling into the trap

The discipline that makes "industry-agnostic" work is sequencing. Each step must earn the next: win the vertical, harvest references + a hardened standards pack, then hop to the nearest ring that reuses the most.

Phase	Target	Why now / what it proves	Reused vs new
Now	Pharma + adjacent GxP (med device, nutraceutical) — SMB / emerging-market wedge	Highest pull; the wedge incumbents ignore; proves the engine + reproducible AI + affordability	Baseline
Next	Food & Beverage (Ring 1)	Near-identical supplier-audit + recall + lot-trace; strong budget growth; first proof of "agnostic"	~75% reuse / new HACCP pack
Then	Cosmetics, Blood/Tissue, Cell & Gene (Ring 1)	Same GxP-lite + custody motion; custody is a native strength	~70% reuse
Later	Automotive / Aerospace supplier-audit angle (Ring 2)	Enter via the supplier-audit-network wedge, not full IATF/AS9100 EQMS; ride ISO 9001:2026 convergence	~50% reuse / PPAP, FAI packs
Horizon	Chemicals, Cannabis, Agri, Energy (Ring 3)	Larger config lift; pursue where a specific buyer or partner pulls you in	~40% reuse

The rule that avoids the horizontal trap: never enter a ring on ambition alone. Enter only when (1) the engine reuses enough that the new-config cost is small, (2) a real buyer or partner is pulling you in (not a spreadsheet TAM), and (3) you can win or partner past the resident specialist. "Industry-agnostic" is the *destination* reached one earned vertical at a time — not the opening pitch.

The wedge that travels across every ring: supplier audit. Notice that supplier qualification + audit is the one use case present in *every* sector (pharma, food, auto, aero, electronics, chemicals). It is the most industry-portable entry point and the one with the worst incumbent coverage (audit-network players are pharma-only; EQMS players don't run networks). Leading each new vertical with the supplier-audit wedge — rather than the full EQMS — is the lowest-config, highest-pain way to enter, then expand into internal quality once inside.

What would have to be true — and what could kill it

The risks, named honestly

Risk	Mitigation / what must be true
The horizontal trap — generalist loses to every specialist	Sequence vertically; win + reference one ring before the next; keep per-vertical depth real, not cosmetic.
Config cost is higher than hoped — each vertical is secretly bespoke	Be ruthless that standards/vocab/rules are truly data, not code. If a new vertical needs core changes, the architecture is leaking.
Horizontal platforms (ServiceNow, SAP, Salesforce, GRC) claim the same primitive	They have distribution; you must have regulated-domain depth + reproducible AI they lack. Compete on GxP credibility, not platform breadth.
Beachhead too small to fund expansion	The SMB/emerging-market wedge must generate enough revenue + references to fund Ring-1 entry. Watch this closely.
Two-sided audit-network liquidity (cold-start)	Use the SaaS base's captive suppliers as initial supply; don't depend on the network for the early value prop.
Per-vertical validation + references don't transfer	True — each regulated vertical needs its own proof. Budget for it; treat the standards pack + a reference customer as the real cost of a new ring.

The verdict. Your instinct is right on two counts and needs discipline on a third. **Right (1):** competing with pharma EQMS incumbents head-on is the wrong game — they own vertical depth and are closing their own white spaces; treat them as not-your-endgame. **Right (2):** the genuinely unoccupied space is the industry-agnostic compliance engine, and the standards world is converging toward exactly that architecture — the tailwind is real. **The discipline (3):** "industry-agnostic" wins only as *sequenced verticalization* — beachhead deep, then ring by ring by architectural distance, led by the supplier-audit wedge that travels everywhere — never as a horizontal pitch from day one. The framework applies to a dozen industries; the strategy is to earn them one at a time.

Honesty & sourcing. Market figures are from third-party research firms (Fortune Business Insights, Grand View, Mordor, Fact.MR, Verdantix, Emergen, 2025–26) and vary by definition; ranges shown, not summed. Sector fit/pull/density scores are analytical judgments from the standards landscape and competitive research, not surveyed data — validate each sector with real buyer discovery before committing GTM spend. Incumbent named only to mark categories, not to rank. The convergence of ISO 9001:2026 / IA9100 / IATF 16949:2027 is sourced to industry reporting and should be confirmed against the standards bodies. This is a strategy analysis, not a forecast.